

Engine



Special Tool(s)

 ST3101-A	Bar, Engine Spreader 303-1454
 ST1341-A	2,200# Floor Crane, Fold Away 300-OTC1819E or equivalent
 ST1636-A	Retainer, Torque Converter 307-346 (T97T-7902-A) or equivalent

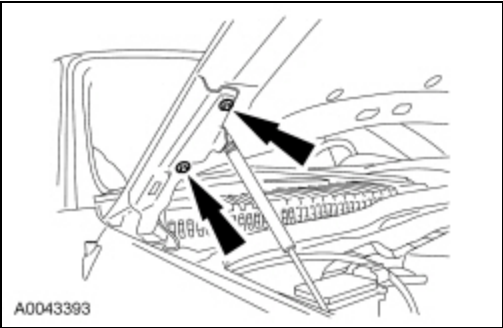
WARNING: Do not smoke, carry lighted tobacco or have an open flame of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.

WARNING: Before working on or disconnecting any of the fuel tubes or fuel system components, relieve the fuel system pressure to prevent accidental spraying of fuel. Fuel in the fuel system remains under high pressure, even when the engine is not running. Failure to follow this instruction may result in serious personal injury.

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to Section 100-02.
2. **NOTE:** Index-mark the hood hinge location to aid in hood installation.

NOTE: RH shown, LH similar.

Remove the 4 bolts (2 shown) and the hood.

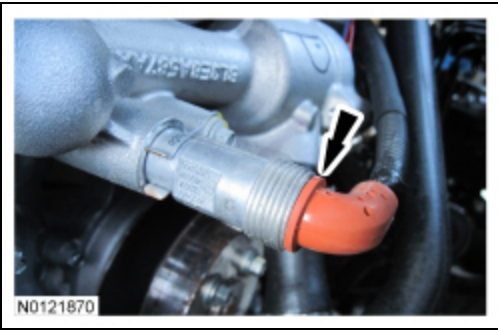


3. Release the fuel system pressure. For additional information, refer to Section 310-00.

4. Disconnect the battery ground cable. For additional information, refer to Section 414-01.

5. If equipped, disconnect the block heater electrical connector at the thermostat housing.

- Detach the block heater wiring harness retainer from the LH valve cover stud bolt and position aside.

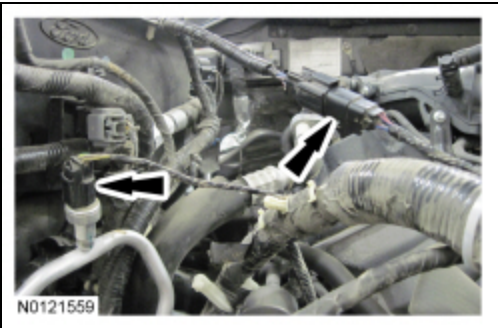


6. Remove the cooling module. For additional information, refer Section 303-03.

7. Remove the LH and RH turbocharger intake tubes. For additional information, refer to 3.5L Gasoline Turbocharged Direct Injection (GTDI) Charge Air Cooler (CAC) and Tubes Exploded View in Section 303-12.

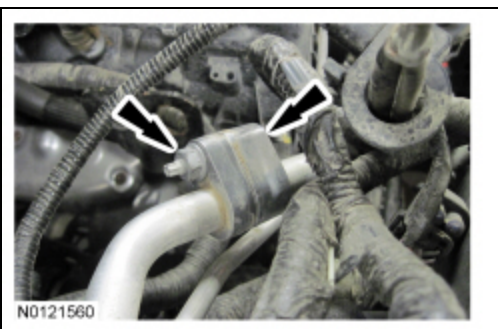
8. Remove the 2 Charge Air Cooler (CAC) inlet tubes and the CAC outlet tube. For additional information, refer to 3.5L Gasoline Turbocharged Direct Injection (GTDI) Charge Air Cooler (CAC) and Tubes Exploded View in Section 303-12.

9. Disconnect the Battery Junction Box (BJB) wiring harness and the A/C pressure transducer electrical connectors.

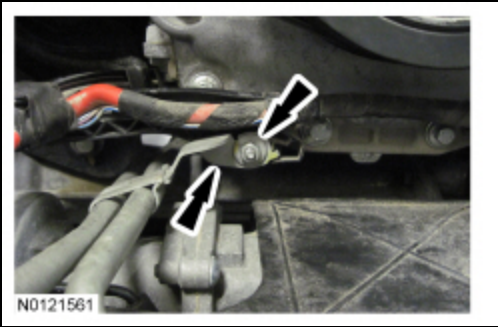


10. Remove the nut and disconnect the A/C tube.

- Discard the O-ring seal and gasket seal.



11. Remove the nut and the transmission cooler line bracket from the engine front cover stud bolt.

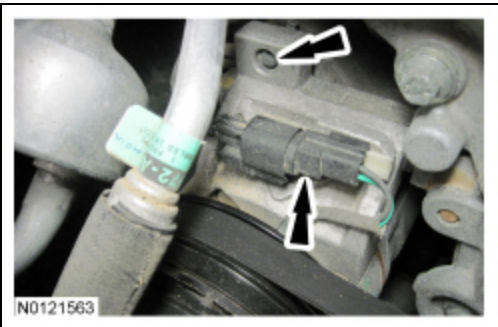


12. Detach the generator wiring harness from the engine front cover.



13. Disconnect the A/C compressor electrical connector.

- Detach the A/C compressor wiring harness from the A/C compressor.

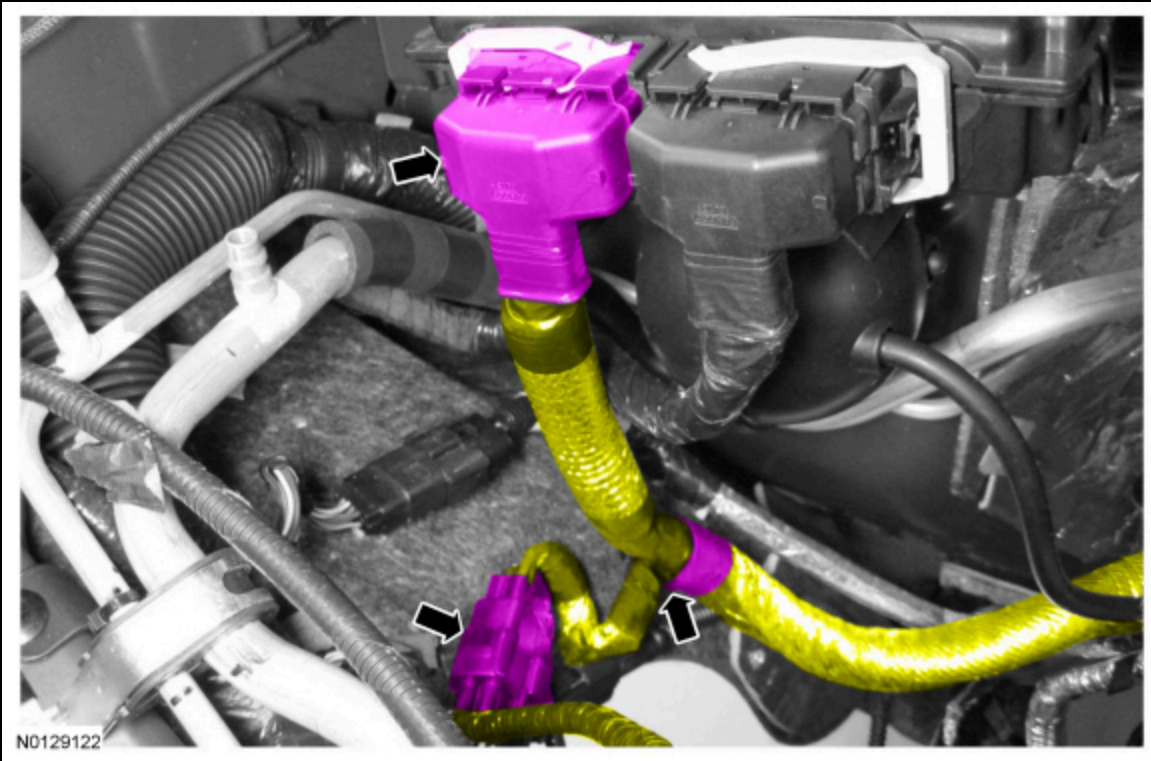


14. Remove the generator. For additional information, refer to Section 414-00.

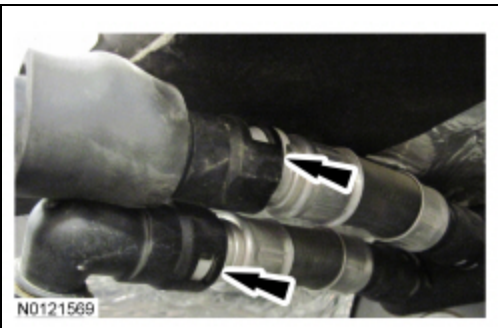
15. Remove the A/C compressor belt. For additional information, refer to Section 303-05.

16. Disconnect the PCM electrical connector and the engine wire harness electrical connector.

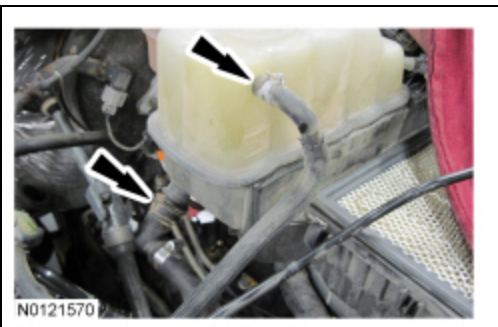
- Detach the wiring harness retainers from the fender splash shield.



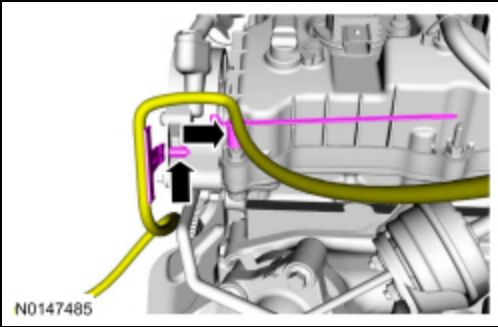
17. Remove the fuel injection pump. For additional information, refer to Section 303-04A.
18. Remove the intake manifold. For additional information, refer to Intake Manifold in this section.
19. Disconnect the 2 heater hose couplings at the RH rear of the engine.



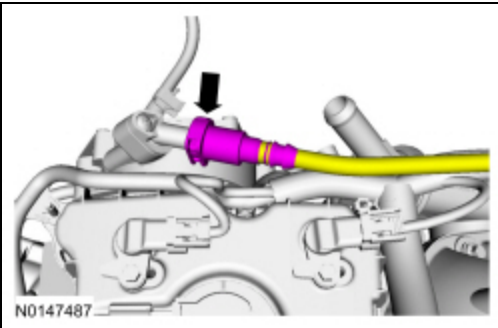
20. Disconnect the 2 coolant hoses from the degas bottle.



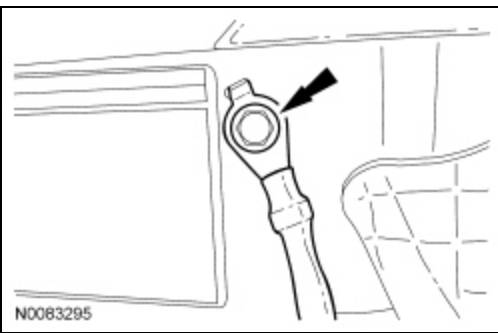
21. Detach the wiring harness retainers from the rear of the RH valve cover stud bolt and the rear of the vacuum pump.



22. Disconnect the vacuum pump tube and position aside.



23. Remove the bolt and the ground strap from the cowl.



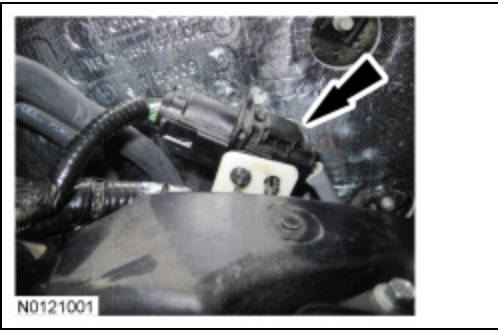
24. Remove the cowl panel grille. For additional information, refer to Section 501-02.

25. Remove the 6 screws and release the 2 cowl panel clips and remove the cowl panel.

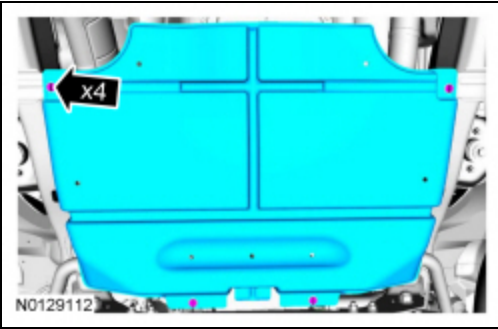


26. Disconnect the Heated Oxygen Sensor (HO2S) electrical connector from the engine wiring harness at the rear of the LH cylinder head.

- Disconnect the HO2S electrical connector from the wiring harness retainer.



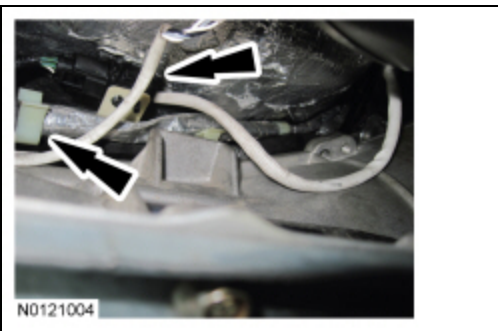
27. Release the 4 fasteners and the splash shield.



28. **NOTE:** Only one wire harness retainer and one electrical connector shown, others similar.

Disconnect the 2 HO2S electrical connectors.

- Disconnect the 2 HO2S electrical connector retainers.
- If equipped, disconnect the 5 transmission wire harness retainers from the transfer case wire harness.



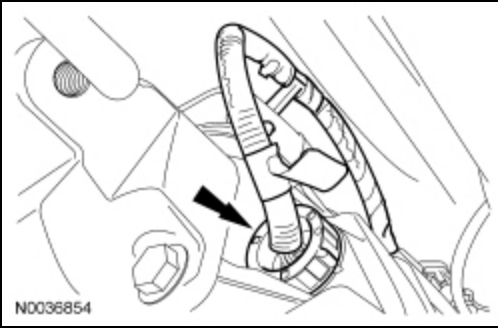
29. Disconnect the HO2S electrical connector at the LH rear of the transmission.

- Disconnect the electrical connector retainer from the bracket at the rear of the transmission.



30. **NOTICE:** Do not pull on the wire harness to disconnect the connector or damage to the connector will occur.

Disconnect the transmission vehicle harness connector by twisting the outer shell and pulling back on the connector.

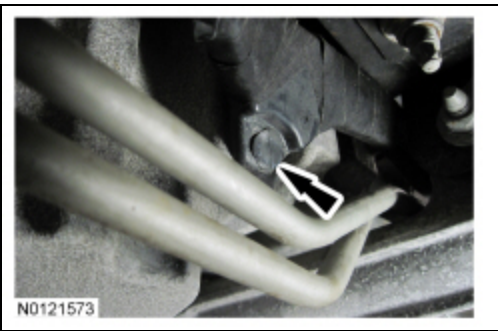


31. Remove the 4 bolts and the skid plate.

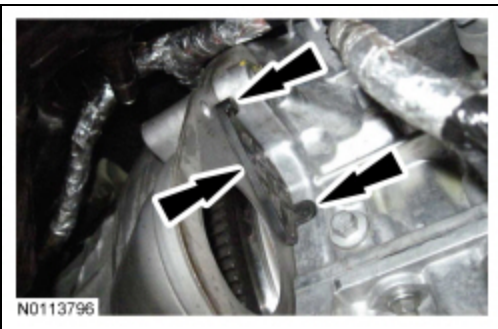
32. If equipped, remove the front driveshaft. For additional information, refer to Section 205-01.

33. Remove the starter motor. For additional information, refer to Section 303-06.

34. Detach the starter motor wiring harness retainer from the oil pan.



35. Remove the 2 pin-type retainers and the access cover.



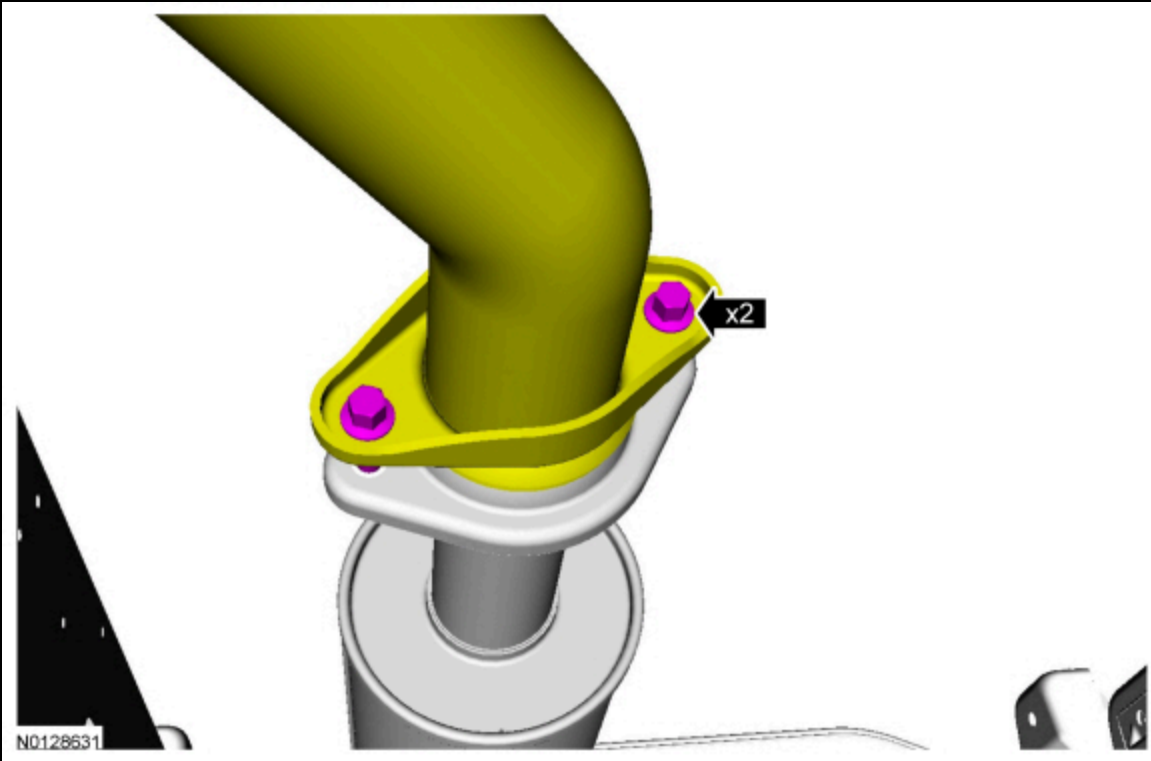
36. **NOTE:** Index-mark the end of one torque converter stud and the flexplate for installation.

Remove the 4 torque converter nuts.



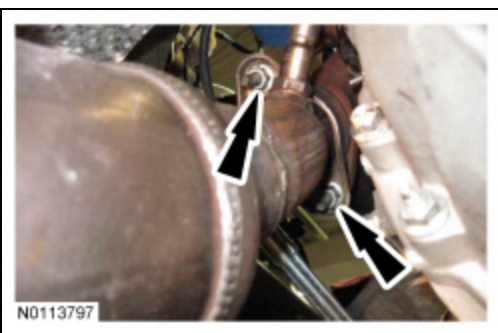
37. Remove the 2 exhaust Y-pipe dual catalytic converter-to-exhaust intermediate pipe bolts and separate.

- Discard the bolts.



38. **NOTE:** *LH shown, RH similar.*

Remove and discard the 2 LH and RH catalytic converter-to-exhaust manifold nuts.



39. Remove the drain plug and drain the engine oil.
- Install the drain plug and tighten to 27 Nm (20 lb-ft).

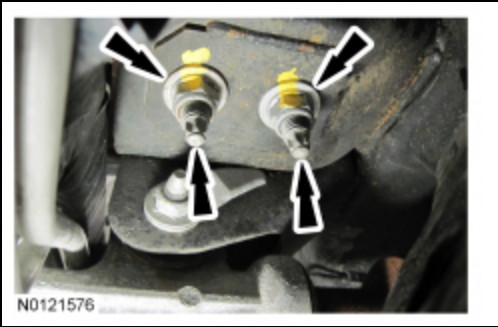


40. Remove and discard the engine oil filter.

41. **NOTICE:** Only use hand tools when removing the engine support insulator nuts and studs or damage to the engine support insulator can occur.

NOTE: *The 2 engine support insulator studs may come off with the nuts.*

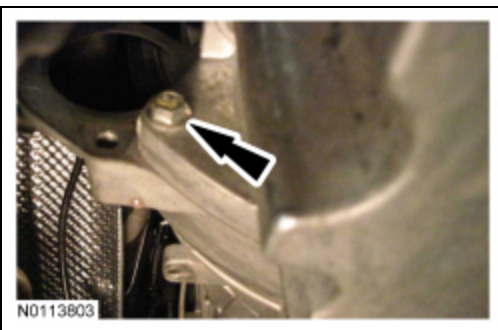
Remove and discard the 2 RH engine support insulator nuts. Remove the 2 engine support insulator studs.



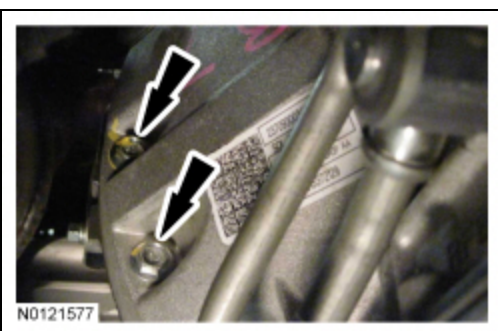
42. Remove the engine-to-bellhousing bolt.



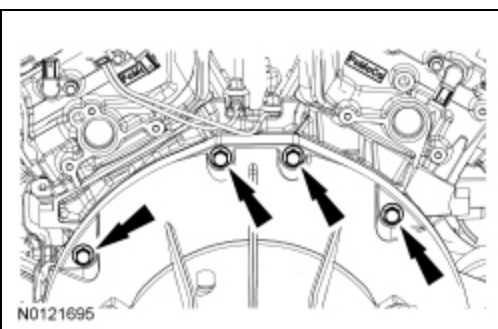
43. Remove the oil pan-to-bellhousing bolt.



44. Remove the 2 LH bellhousing-to-engine bolts.

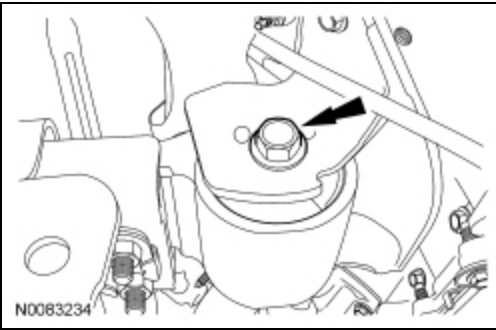


45. Remove the 4 upper bellhousing-to-engine bolts.



46. **NOTICE:** Only use hand tools when loosening or tightening the engine support insulator through bolts or damage to the engine support insulator-to-cylinder block bracket can occur.

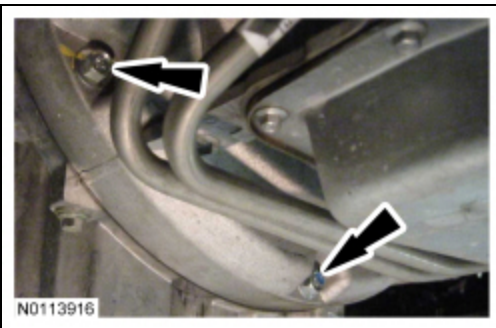
Remove and discard the LH engine mount through bolt.



47. **NOTICE:** Do not support the transmission by the fluid pan, failure to follow this instruction may result in serious damage to the transmission.

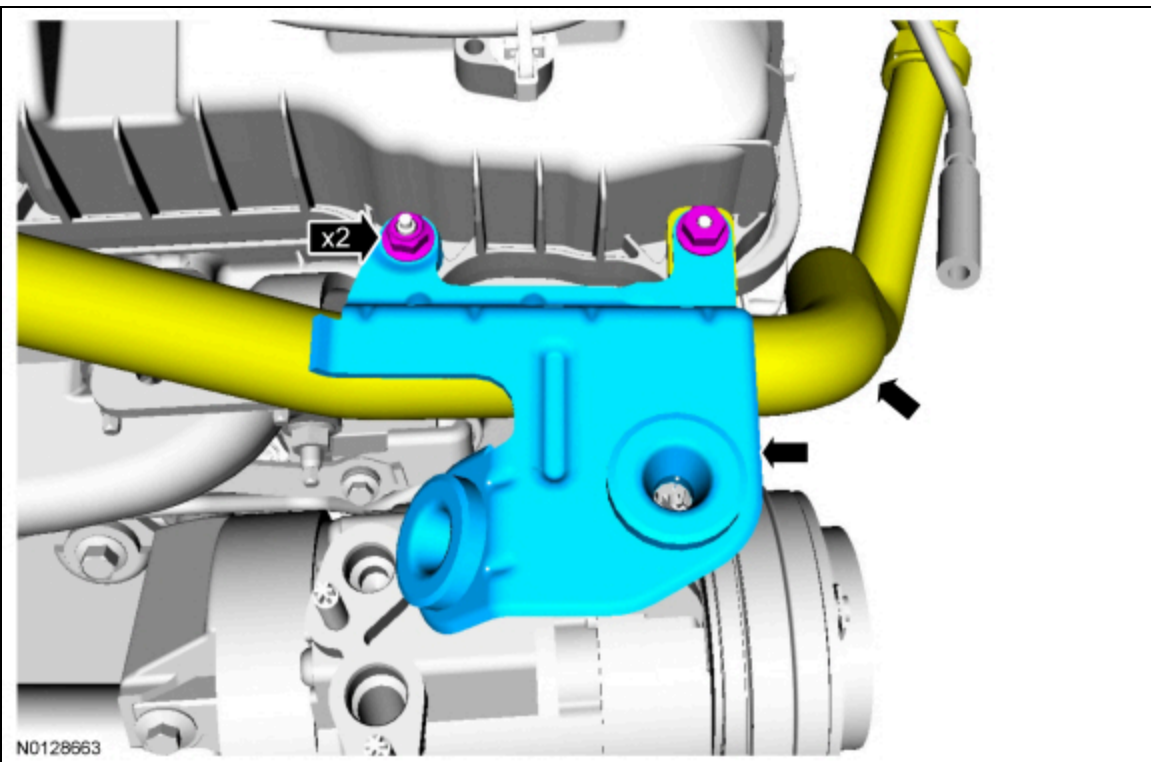
Support the bellhousing of the transmission with a suitable floor jack and a block of wood.

48. Remove the 2 LH bellhousing-to-oil pan bolts.



49. Remove the 2 nuts and the CAC tube bracket.

- Disconnect the coolant hose from the valve cover stud.



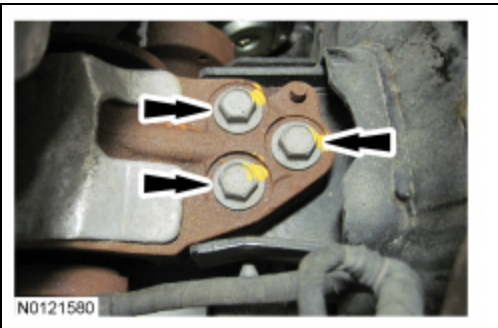
50. Using a piece of chain and 2 quick links, assemble the Engine Spreader Bar as shown.



51. **NOTICE:** Only use hand tools when removing or installing the engine support insulator-to-frame bolts or damage to the engine support insulator-to-frame nut plate can occur.

NOTE: After removing the bolts leave the LH engine support insulator in place.

Remove and discard the 3 bolts for the LH engine support insulator.



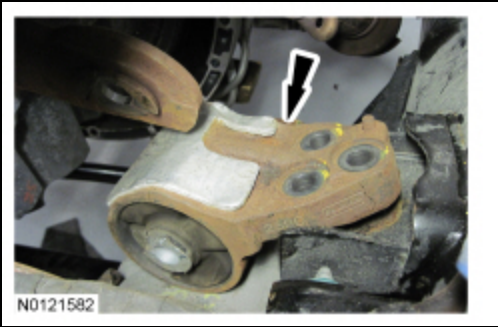
52. **NOTE:** The use of a ratchet strap may be needed to level the engine.

NOTE: Use a commercially available quick link on the right side between the lifting eye and the engine spreader bar.

Using the Floor Crane and Engine Spreader Bar assembly, lift the engine.



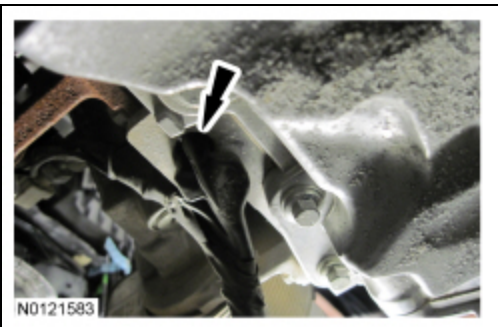
53. Remove the LH engine support insulator.



54. **NOTE:** *The transmission wiring harness is part of the engine wiring harness.*

Detach the generator wiring harness retainer from the lower LH side of the engine block.

- Position the transmission wiring harness on top of the engine for removal.



55. Using the Floor Crane and Engine Spreader Bar assembly, remove the engine.



56. Install the Torque Converter Retainer.

